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Original article

Epidemiology of optic disc edema in 2021/2022: Results from a cohort of 197 patients

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ABSTRACT

Objective. – The aim of our study was to determine the etiologies of optic disc edema between 2021 and 2022.
Materials and methods. – This was a multicentric study at the Timone and Nord university hospitals in Marseille. Patients were retrospectively followed in ophthalmology departments, with inclusion between January 2021 and December 2022. All patients presenting with newly diagnosed uni- or bilateral optic disc edema, both adults and children, were included. Their ophthalmological evaluation included a fundus examination and optical coherence tomography if feasible.
Results. – In total, 197 patients were included. Intracranial hypertension (IH) was the most frequent etiology (37.06%). The primary causes of IH were idiopathic (27/73), intracranial tumors (21/73), and cerebral venous thrombosis (12/73). The second etiology of optic disc edema was retinal vein occlusion in 19.9% of cases (39/197). Edema reactive to uveitis was found in 13.2% of cases (26/197). Finally, inflammatory (17/197) and ischemic (30/197) optic neuropathies were identified.
Conclusion. – This study updates the most frequent etiologies of optic disc edema in 2021 and 2022 to facilitate diagnostic hypotheses for de novo optic disc edema. It highlights the importance of a comprehensive and personalized evaluation in diagnosing optic disc edema, taking into account recent advances in imaging techniques and biomarkers.

1. Introduction

Optic disc edema is characterized by swelling of the neuroretinal rim, resulting in elevation and blurred margins of the optic disc at fundus examination. These are significant clinical indicators of various ophthalmological, neurological, and systemic pathologies. With an aging population and the rise of neurological diseases, understanding these indicators has become more relevant than ever. The diagnosis and understanding of optic disc edema have long been limited by the available diagnostic tools. However, with the advent of

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advanced techniques such as optical coherence tomography (OCT) and improved imaging methods, it is now possible to identify underlying causes with greater accuracy [1]. These diagnostic advancements play a crucial role in guiding appropriate therapeutic strategies. In this context, this study aimed to determine the etiologies of optic disc edema in 2021 and 2022, a period marked by significant evolution in knowledge and diagnostic capabilities. This research contributes to the ongoing improvement in patient care, offering an updated perspective on the causes of optic disc edema.

2. Materials and methods

We retrospectively reviewed the clinical records of patients presenting with optic disc edema at the Timone and Nord University Hospitals in Marseille, between January 2021 and December 2022. All patients, both adults and children, presenting with uni- or bilateral optic disc edema were included.

The ophthalmological data collected included fundus examination and OCT findings when available. The following OCT data were collected: retinal nerve fiber layer (RNFL) thickness, total peripapillary thickness, and the ganglion cell layer (GCL) primarily for optic disc edema quantification.

The OCT used was a Heidelberg Spectralis SDOCT, Heidelberg Engineering, Germany, software version Spectralis 5.3.3.0, Eye Explorer software 1.6.4.0.

Table 1 – Demographic and clinical characteristics of the cohort.

Cohort, *n*	197 patients
Age, years	41.48 (3.92–96.16)
Mean (range)	
Male sex, *n* (%)	102 (51.78%)
Unilateral involvement, *n* (%)	101 (51.27%)

The following clinical data were collected: age, sex, and etiology of the optic disc edema.

Patients with optic neuritis or intracranial hypertension underwent brain imaging. For patients with ischemic optic neuropathies, angiography was performed as part of the etiological assessment.

3. Results

Data from 197 patients were collected, with their demographic and clinical characteristics detailed in Table 1.

The average visual acuity at diagnosis was 0.43 ± 0.81 (logMAR) in the total cohort and 0.19 ± 0.45 (logMAR) in children under 15 years old.

Analysis of the etiologies of optic disc edema revealed a diversity of causes. The predominant etiology was intracranial hypertension (IH), accounting for 73 of the 197 cases. This was closely followed by retinal vein occlusions (RVO), affecting

Table 2 – Etiologies and visual acuities of optic disc edema and their distribution in children under 15 years old.

Etiologies	Total (%) *Unilateral* *Bilateral*	Visual acuity (logmar) Mean ± SD	Children < 15 y.o.	Visual acuity of children < 15 y.o.
Intracranial hypertension	73 (37.06%) *6* *67*	0.12 ± 0.41	18	0.11 ± 0.27
Retinal vein occlusion	39 (19.8%) *0* *39*	0.78 ± 0.93	0	
Acute anterior ischemic optic neuropathy	30 (15.23%) *26* *4*	1.04 ± 1.21	0	
Uveitis	26 (13.2%) *8* *18*	0.28 ± 0.52	10	0.18 ± 0.32
Inflammatory optic neuropathy	17 (8.63%) *16* *1*	0.93 ± 1.03	3	0.22 ± 0.32
Compression	5 (2.54%) *5* *0*	0.88 ± 1.17	0	
Malignant hypertension	3 (1.52%) *1* *2*	0.009 ± 0.02	0	
Leukemia	2 (1.01%) *0* *2*	0 ± 0	2	0 ± 0
Iatrogenic	1 (0.51%) *1* *0*	0	0	
Trauma	1 (0.51%) *0* *1*	2.7	1	2.7

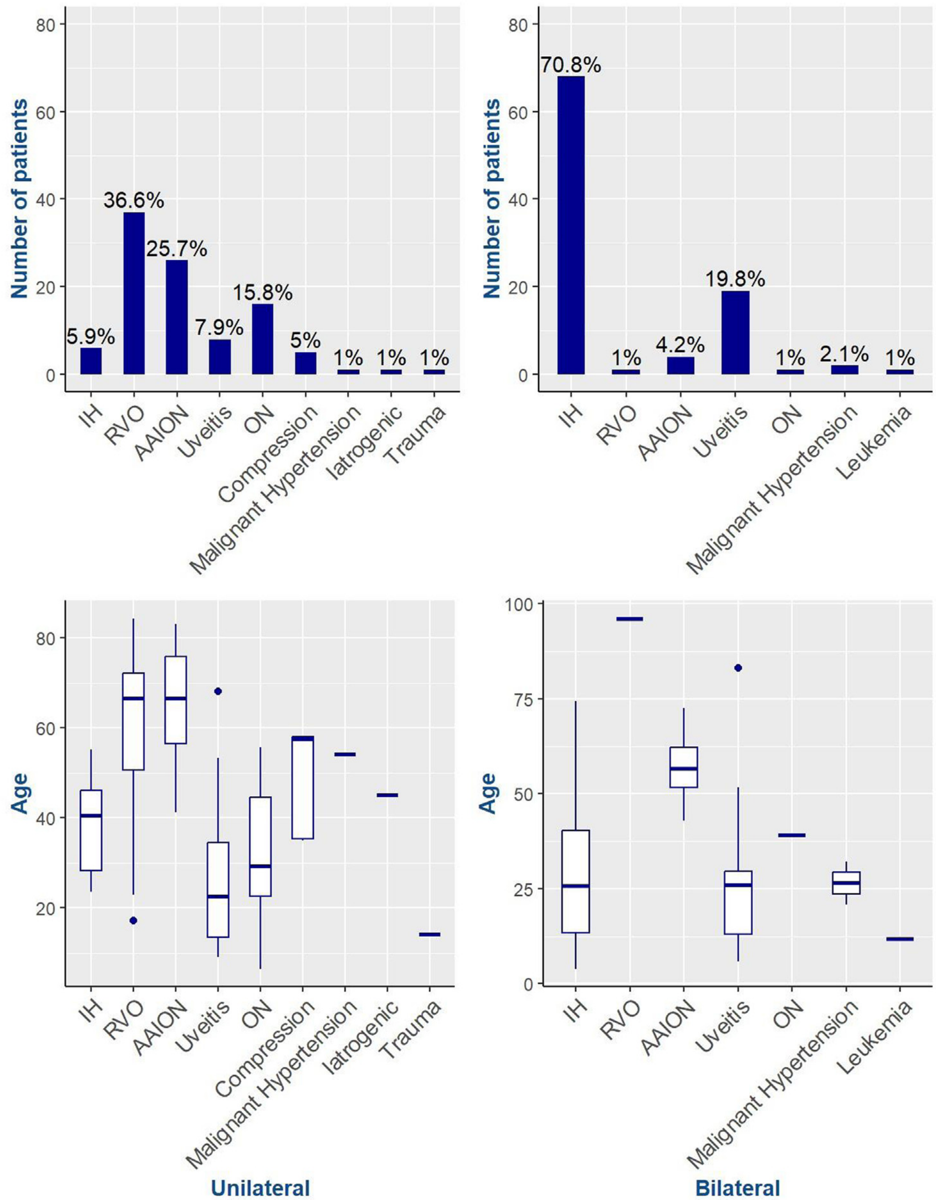

Fig. 1 – Etiologies of optic disc edema according to unilateral (A) or bilateral (B) nature and the age distribution of etiologies based on unilateral (C) or bilateral (D) nature. IH: intracranial hypertension; RVO: retinal vein occlusion; AAION: acute anterior ischemic optic neuropathy; ON: inflammatory optic neuropathy.

39 patients, and acute anterior ischemic optic neuropathy (AAION), observed in 30 cases. We also identified 26 cases of uveitis and 17 cases of inflammatory optic neuritis (ION). The less common etiologies and the uni- or bilateral nature of optic disc edema were detailed in Table 2. Out of 197 patients, 34 were children under 15 years old, and the various etiologies are also detailed in Table 2.

In Fig. 1A and B, we presented the different etiologies of optic disc edema based on whether the condition was unilateral or bilateral. When the condition was bilateral (Fig. 1B), the most common etiology was intracranial hypertension (70.8%), followed by uveitis (19.8%). When the condition was unilateral (Fig. 1A), the most common etiologies were retinal vein occlusions (36.6%), anterior ischemic optic neuropathies (25.7%), and inflammatory optic neuritis (15.8%).

In Fig. 1C and D, we present the age distribution of patients according to the different etiologies and whether the optic disc edema was unilateral or bilateral. We observed that intracranial hypertension, uveitis, and inflammatory optic neuritis were more common in younger patients, while retinal vein occlusions and anterior ischemic optic neuropathies were more frequently seen in the older population.

As for IH, the main causes were idiopathic IH (27 out of 73 cases), tumor etiologies (21 out of 73), and cerebral venous thrombosis (12 out of 73). Other less common causes were identified, such as meningitis (5 out of 73), cerebral hematomas or hemorrhages (spontaneous or traumatic) (5 out of 73), stenosis of the aqueduct of Sylvius (2 out of 73), and an arteriovenous malformation (1 out of 73).

Regarding AAION, the majority was non-arteritic, representing 83.33% (25 out of 30 cases), while 16.67% (5 out of 30) were arteritic.

For inflammatory optic neuritis, multiple sclerosis was the most frequently observed etiology, accounting for 35.29% (6 out of 17) of cases. MOG antibody-associated neuritis (MOGAD) and idiopathic neuritis each represented 23.53% (4 out of 17) of cases. More rarely, we encountered cases of neuritis secondary to sarcoidosis (2 out of 17) and acute disseminated encephalomyelitis (1 out of 17).

4. Discussion

This study provides an update on the most frequent etiologies of optic disc edema at the current state of knowledge. Very few studies have sought to determine the causes of optic nerve head edema. The team of Sachdeva et al. reported findings of papilledema in emergency ophthalmology settings: one in two patients presenting with papilledema had IH [2].

The diversity of etiologies observed in our study reflects the complexity of diagnosing optic disc edema. The high prevalence of IH aligns with trends observed in previous studies [3], but the low proportion of idiopathic IH in our cohort may be explained by the fact that our university hospital center is a regional reference center for neurovascular and neurosurgical emergencies. Therefore, tumor and neurovascular etiologies are particularly high in this study. This observation underscores the importance of the clinical context in the epidemiology of optic disc edema.

Furthermore, it is important to note the recent increase in the incidence of idiopathic IH [4]. This rising trend can be attributed to several factors, including the increase in obesity, which is a major risk factor for idiopathic IH. Changes in lifestyle habits, such as sedentary behavior and high-calorie diets, also contribute to this rise. Additionally, the improvement in diagnostic techniques allows for the identification of more cases of idiopathic IH that might have previously gone unnoticed. These elements highlight the need to consider these risk factors in the management and prevention of idiopathic IH.

The average age of the patients highlights a prevalence in the adult population, in line with data from Parajuli et al. [5]. The gender distribution is relatively balanced, which is consistent with the conclusions of Osaguona et al. [6].

About one in four patients in this study with a diagnosis of inflammatory optic neuritis had anti-MOG antibodies. This can be explained on the one hand because it is a newly described entity, first identified in 2014 [7], and whose diagnostic advances are improving day by day. On the other hand, optic disc edema in optic neuritis is found in 30% of multiple sclerosis cases, in more than 90% of MOGAD, and exceptionally in neuromyelitis optic associated with anti-AQP4 antibodies [8,9]. These findings suggest the need for a more nuanced diagnostic approach, combining clinical evaluation and serological tests, for precise identification of underlying causes.

This study highlights the importance of clinical context and presentation in the diagnostic approach. For example, bilateral papilledema in a young patient with relatively preserved visual acuity suggests intracranial hypertension. Bilateral papilledema accompanied by red, painful eyes and decreased visual acuity suggests uveitis. Unilateral papilledema in an elderly individual should raise suspicion of anterior ischemic optic neuropathy or retinal vein occlusion. In young patients, unilateral papilledema is more indicative of inflammatory optic neuritis or uveitis.

The limitations of this study include the sample size, which, although substantial, may not be representative of the general population. Moreover, the lack of long-term follow-up limits our ability to understand the long-term outcomes and evolutionary trends of optic disc edema. Future studies should aim to include a larger number of participants and provide follow-up over a more extended period.

5. Conclusion

This study contributes to the expansion of knowledge on the etiologies of optic disc edema within a contemporary clinical framework. It highlights the importance of a comprehensive and personalized evaluation in the diagnosis of optic disc edema, taking into account recent advancements in imaging techniques and biomarkers. These results underscore the need for close collaboration between ophthalmologists and neurologists for the optimal management of patients.

Disclosure of interest

The authors declare that they have no competing interest.

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